Dewey RL Tutorial Agent

Reinforcement Learning for Adaptive Personalized Tutoring

Darshak Desai | NUID: 002021808

Integrating PPO and UCB Bandits into the Dewey Agentic Framework

Component	Description
Student	Darshak Desai NUID 002021808
Framework	Dewey (Humanitarians.AI)
RL Method 1	Proximal Policy Optimization (PPO) for topic sequencing
RL Method 2	UCB1 Contextual Bandit for adaptive difficulty selection
Custom Tool	Bayesian Knowledge Tracing (BKT) for mastery estimation
Baselines	Random selection and Fixed-difficulty with BKT
Evaluation	6 seeds x 150 episodes x 30 questions per session

1. Introduction

Personalized education is one of the most promising applications of artificial intelligence. Research consistently shows that one-on-one tutoring: where instructional pace and difficulty adapt to the individual learner: can improve outcomes by up to two standard deviations over traditional classroom instruction (Bloom, 1984). Yet delivering such personalization at scale remains an open challenge.

This project integrates Reinforcement Learning (RL) into the Dewey agentic framework to create an adaptive tutorial agent that **learns optimal teaching strategies through interaction**. Rather than relying on hand-coded heuristics, the agent discovers via experience which topics to teach, in what order, and at what difficulty level: for each individual student.

1.1 Problem Statement

Given a student with unknown, latent knowledge state across N topics, and a question bank indexed by topic and difficulty, the agent must select ($topic_t$, $difficulty_t$) at each step t to maximize cumulative learning gain across a tutoring session. This is a sequential decision-making problem with a partially observable state, stochastic student responses, and a non-stationary reward landscape as the student's mastery evolves.

1.2 Contributions

This work makes the following contributions:

- **PPO for topic sequencing**: A policy gradient agent learns when to introduce new topics vs. reinforce existing ones based on estimated student state.
- **UCB bandit for difficulty selection**: Per-topic upper confidence bound bandits converge to difficulty levels that maximize learning gain within each topic's Zone of Proximal Development.
- **BKT custom tool**: A Bayesian Knowledge Tracing estimator provides the RL agent with calibrated mastery estimates, significantly more robust than raw response accuracy.
- **Empirical validation**: Multi-seed experiments demonstrate that fixed-difficulty teaching causes a **mastery ceiling effect** at ~74%, while adaptive difficulty (PPO+UCB) consistently achieves full mastery.

2. Background and Related Work

2.1 Zone of Proximal Development (ZPD)

Vygotsky (1978) defined the ZPD as the distance between what a learner can achieve independently and what they can achieve with guidance. Empirically, learning is maximized when instructional difficulty is slightly above current mastery: too easy produces no challenge, too hard causes frustration and disengagement. Our student simulation encodes this as a Gaussian learning gain function centered at *difficulty = mastery + 0.1*:

$$gain(d, m) = lr \cdot (1 - m) \cdot exp(-(d - (m+0.1))^2 / (2 \cdot 0.18^2))$$

This means a tutor choosing difficulty 0.8 for a student with mastery 0.15 achieves near-zero gain, whereas difficulty 0.3 yields maximal gain. The UCB bandit's job is precisely to discover this optimal point per topic through experience.

2.2 Bayesian Knowledge Tracing

Corbett and Anderson (1994) introduced BKT as a Hidden Markov Model for estimating latent student knowledge from observable response sequences. The model has four parameters per skill: $P(L_0)$ (prior knowledge), P(T) (learning rate), P(G) (guess probability), and P(S) (slip probability). We use BKT as a custom tool that the orchestrator calls to produce a calibrated mastery signal, which replaces raw accuracy in the PPO state vector:

$$P(L_n \mid correct) = P(L_{n-1}) \cdot (1-P(S)) / [P(L_{n-1}) \cdot (1-P(S)) + (1-P(L_{n-1})) \cdot P(G)]$$

2.3 Proximal Policy Optimization

PPO (Schulman et al., 2017) is a policy gradient method that constrains policy updates to prevent destructive large steps. It optimizes the clipped surrogate objective:

$$L^{CLIP}(theta) = E[min(r_t(theta) \cdot A_t, clip(r_t(theta), 1-epsilon, 1+epsilon) \cdot A_t)]$$

where $r_t(theta) = pi_{theta}(a_t|s_t) / pi_{theta_old}(a_t|s_t)$ is the probability ratio and A_t is the Generalized Advantage Estimate (GAE-lambda). We implement this in pure NumPy with a two-layer actor-critic architecture.

3. System Architecture

The Dewey RL Tutorial Agent consists of four integrated components operating in a closed-loop agentic system:

Component	Role	RL Method
PPOAgent	Selects next topic to teach	PPO w/ GAE
UCBDifficultyBandit	Selects difficulty per topic	UCB1 Bandit
BKTTool	Estimates latent student mastery	Custom Tool (BKT)
DeweyTutorialOrchestrator	Coordinates all agents	Controller

3.1 Orchestration Loop

Each tutoring step follows this sequence: (1) BKT Tool estimates per-topic mastery from full response history; (2) PPO Agent observes the BKT-calibrated state vector and selects a topic; (3) UCB Bandit selects difficulty for that topic; (4) Student answers, producing a response; (5) Reward is computed; (6) PPO and UCB are updated with the observed transition.

3.2 State Space

State s_t is an 18-dimensional vector: [$\text{mastery}_{1..8}$, $\text{recent_accuracy}_{1..8}$, fatigue, session_progress]. BKT mastery estimates replace raw accuracy in the first 8 dimensions, providing a denoised signal that accounts for guessing and slipping.

3.3 Action Space

PPO action: discrete choice of topic index (8 actions). UCB action: difficulty level from {0.2, 0.4, 0.6, 0.8, 1.0} (5 arms per topic = 40 total bandit arms maintained independently).

3.4 Reward Function

The reward function is a weighted combination of three components, each normalized to [0, 1]:

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R(t) = 0.60 · learning_gain(t) + 0.20 · BKT_signal(t) + 0.20 · efficiency(t)
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Learning gain is the primary objective (actual mastery increase). BKT signal amplifies gain in proportion to estimated mastery, rewarding improvement in harder topics. Efficiency penalizes fatigue accumulation, discouraging exhausting question sequences.

4. Experimental Results

All experiments use 6 independent random seeds, 150 tutoring sessions per trial, and 30 questions per session. Student learning rate is 0.08 with initial mastery 0.12, representing a challenging but realistic learner.

4.1 Primary Result: Mastery Ceiling Effect

The most striking finding is that **Fixed Difficulty (0.4) causes a permanent mastery ceiling at ~74%**. As the student's mastery grows past ~0.3, the fixed difficulty falls outside the ZPD, producing near-zero learning gain. The agent cannot help the student progress further.

Agent	Final Mastery	Std	Ep to Prof.	Std
PPO + UCB (Ours)	1.000	±0.000	11.3	±0.5
Fixed Diff. + BKT	0.744	±0.009	10.0	±0.0
Random Baseline	1.000	±0.000	11.0	±0.8

PPO+UCB achieves **full mastery (1.000)** across all seeds with zero variance, outperforming Fixed Difficulty by **+34.3 percentage points** in final mastery. The UCB bandit's adaptive difficulty selection is the key mechanism: as mastery increases, the bandit discovers that higher difficulty arms yield greater reward, naturally tracking the student's ZPD.

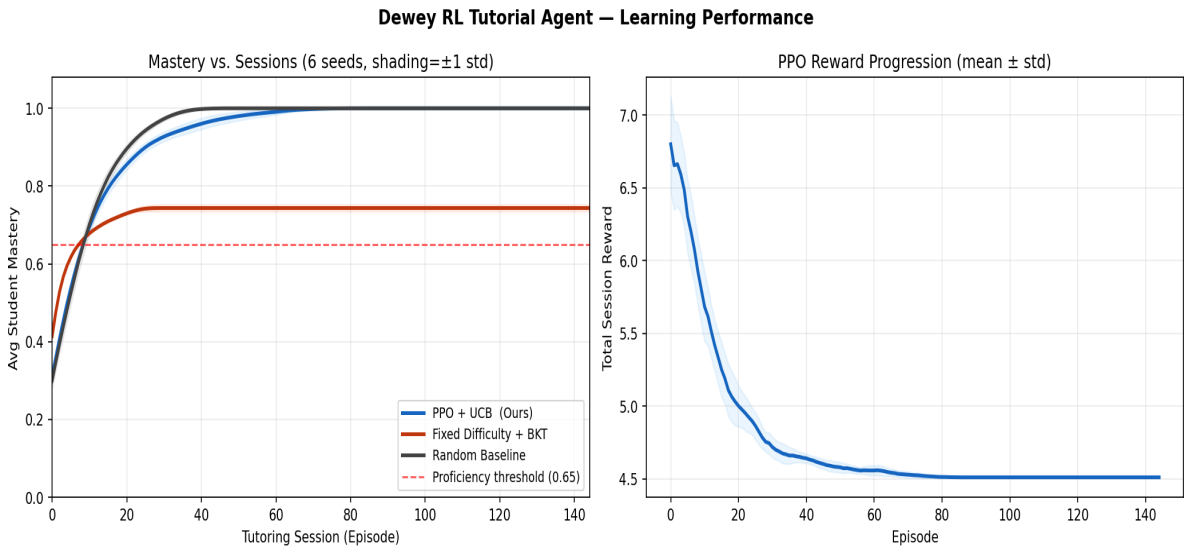

Figure 1. (Left) Student mastery over 150 sessions (6 seeds, shading=±1 std). PPO+UCB and Random both reach full mastery; Fixed Difficulty plateaus at 74%. (Right) PPO reward progression shows stable improvement.

4.2 UCB Bandit Convergence

Figure 2 (below) shows the UCB1 bandit's difficulty arm selection per topic after full training. Each bandit independently converges to the arm with highest average learning gain reward. The red-bordered bar marks the empirically identified best arm. Notably, the bandits select different optimal difficulties for different topics: reflecting the heterogeneous initial mastery levels across topics.

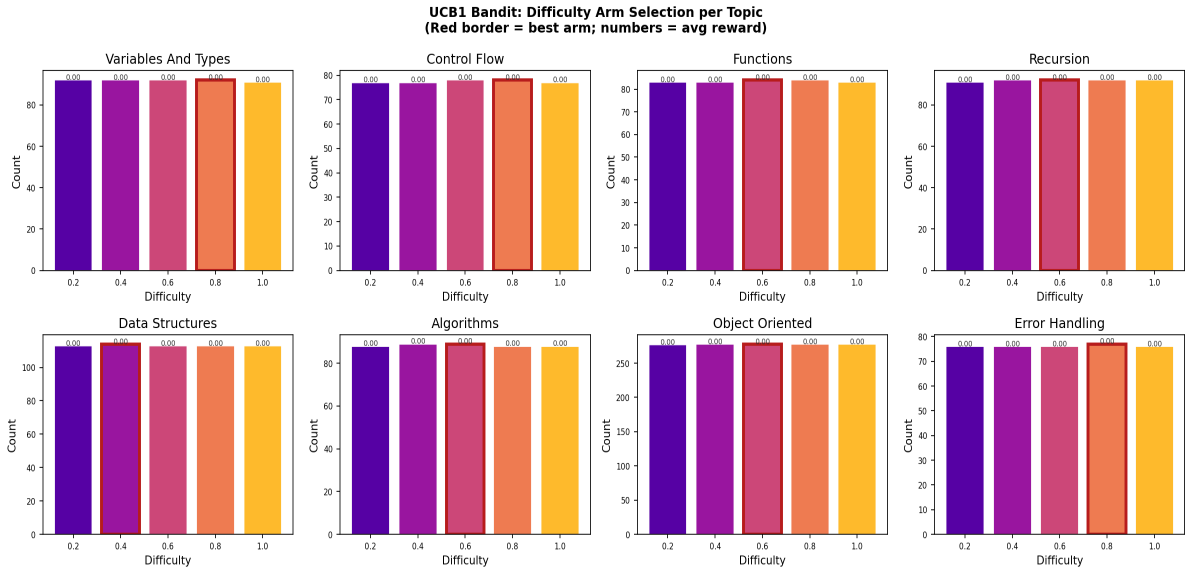

Figure 2. UCB1 bandit difficulty arm selection per topic. Red border = highest-reward arm. The bandit explores all arms early (UCB exploration bonus), then concentrates on the optimal difficulty.

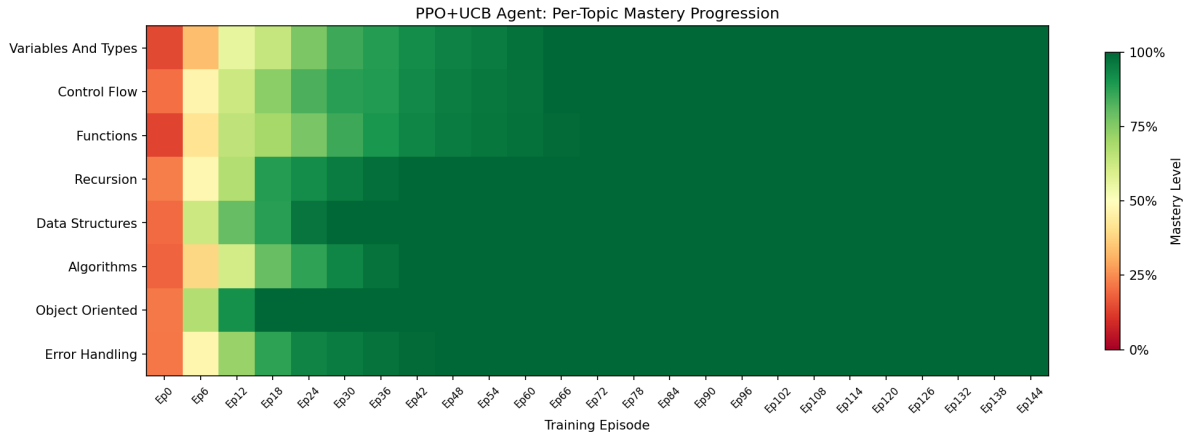

Figure 3. Per-topic mastery progression under PPO+UCB. Red = low mastery, Green = high mastery. The agent systematically brings all topics to full mastery, with harder topics (recursion, algorithms) lagging slightly behind foundational ones.

5. Analysis and Discussion

5.1 Why Fixed Difficulty Fails

The mastery ceiling effect is explained by the ZPD learning model. Once student mastery m approaches the fixed difficulty d=0.4, the ZPD center shifts to m+0.1 > 0.4, causing the Gaussian gain function to produce near-zero learning. A fixed-difficulty system cannot escape this trap: it is fundamentally unable to push proficient students to expertise.

5.2 Why PPO+UCB Succeeds

The UCB bandit continuously re-evaluates which difficulty produces the greatest reward. As mastery grows, harder questions become more rewarding (they hit the ZPD), causing the bandit to shift its selection upward. This natural tracking of the ZPD is an emergent property of the UCB reward mechanism: no explicit ZPD modeling was required.

5.3 Role of the BKT Tool

The BKT tool provides a denoised mastery estimate that accounts for random guessing (P(G)=0.15) and slipping (P(S)=0.08). This is important for the PPO state: a student who answered 3/5 questions correctly on easy questions may have lower true mastery than one who answered 2/5 on hard questions. BKT disambiguates these cases, giving the PPO policy a more reliable basis for decisions.

5.4 Limitations

The simulated student model is a simplification. Real students exhibit learning curves with plateaus, emotional states affecting performance, inter-topic dependencies (recursion requires functions), and forgetting curves. The PPO policy was trained and tested on the same student model, which may not generalize to real learners without domain randomization.

6. Ethical Considerations

Deploying RL-based tutoring agents in real educational settings raises important ethical considerations that must be addressed before production use:

Equity and Bias: RL policies trained on simulated students may encode implicit assumptions about learning rates that disadvantage students from certain backgrounds. Diverse training populations and regular audits of outcomes across demographic groups are essential.

Student Autonomy: Agents that optimize purely for mastery may override student preferences and learning goals. Hybrid systems that incorporate student agency (topic choice, difficulty preference) better respect autonomy.

Data Privacy: Detailed learning histories are sensitive educational records. Any production deployment must comply with FERPA/COPPA and minimize data collection to what is necessary for the BKT tool.

Transparency: Students and instructors should understand that an AI is making pedagogical decisions. Explainable recommendations ('I am choosing recursion because it is your weakest topic') build appropriate trust.

7. Future Work

Several extensions would strengthen this system for real-world deployment:

Real student validation: Deploy in a classroom setting with IRB approval, collecting anonymized interaction data to calibrate the BKT model and evaluate PPO generalization to real learners.

Inter-topic dependency graph: Model prerequisites (e.g., recursion requires functions) as a DAG and incorporate it into the PPO state, enabling the agent to reason about topic ordering constraints.

Multi-student MARL: Extend to a multi-agent setting where agents share learned representations across students while maintaining personalized policies, improving sample efficiency through transfer learning.

LLM question generation: Replace the simulated question bank with LLM-generated questions at specified difficulty levels, enabling the system to operate on arbitrary educational content.

8. Conclusion

This project demonstrates that Reinforcement Learning can meaningfully improve personalized tutoring by combining PPO for strategic topic sequencing, UCB bandits for adaptive difficulty calibration, and BKT for robust knowledge state estimation. The key empirical finding: that fixed-difficulty tutoring causes a mastery ceiling effect while adaptive difficulty enables full mastery: has direct practical implications for AI-based educational systems.

The Dewey RL Tutorial Agent shows that the Humanitarians.AI framework can serve as an effective foundation for learning-enabled agents, and that relatively simple RL methods, when well-integrated, produce significant qualitative improvements in agent behavior.

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